ELECTROLUX ENERGY MANAGEMENT SYSTEM (EMS)

PROJECT PROPOSAL

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1. OVERVIEW

The Electrolux Energy Management System (EMS) is a web-based electricity utility management platform designed to digitize and streamline the operations of electricity distribution companies, with a specific focus on challenges faced by K-Electric customers and staff in Karachi, Pakistan.

The motivation for this project stems from the persistent inefficiencies in how electricity utility services are currently delivered in Pakistan - paper-based billing, manual meter readings, lack of transparent complaint resolution, and no centralized platform for customers to monitor their usage or manage payments.

The system serves three types of users: residential and commercial electricity consumers (customers), field and office staff (employees), and utility company management (administrators). The project will deliver a fully functional, role-based web application covering billing, meter reading management, payment tracking, complaint handling, outage management, and analytical reporting.

The system is being developed by a team of three students over the remaining 8 weeks of the Spring 2026 semester. This project does not depend on any other ongoing projects, nor do any other projects depend on its outcome. It is a self-contained academic and practical deliverable.

2. PROJECT DETAIL

2.1 Problem or Challenge

K-Electric, the sole electricity distribution company serving Karachi, currently relies heavily on manual and paper-based processes for billing, meter reading, and customer service. Customers receive printed bills through physical delivery, have no way to track their consumption history online, and must visit physical offices to register complaints or inquire about payments. This creates significant inconvenience, particularly for working-class citizens and businesses that cannot afford frequent office visits.

Furthermore, the lack of a centralized digital platform leads to administrative inefficiencies on the utility side. Employees manually record meter readings on paper, which introduces human error and delays in bill generation. Complaint resolution has no formal tracking mechanism, leaving customers with no visibility into the status of their issues. Power outage schedules are not proactively communicated, causing disruption to residents and businesses across Karachi. These challenges collectively degrade the quality of service and erode customer trust in the utility company.

2.2 Project Vision

Our vision is to build a transparent, efficient, and accessible digital utility management platform that empowers electricity consumers in Karachi to take control of their accounts while enabling K-Electric staff and administrators to manage operations with greater accuracy and accountability.

The Electrolux Energy Management System aims to eliminate the gap between utility providers and their customers by providing a single, role-based web platform where

customers can view and pay bills, track complaints, and monitor usage - while employees can record meter readings and manage work orders, and administrators can oversee all operations through dashboards and analytics. The ultimate goal is to deliver a solution that mirrors the standards of modern digital utility services seen globally, tailored specifically to the needs and constraints of Karachi's electricity infrastructure.

2.3 Project Feasibility Study

2.3.1 Technical Feasibility

The proposed system is technically feasible. The development team has the required skills in full-stack web development using modern frameworks. The chosen technology stack - Next.js, TypeScript, MySQL, and Drizzle ORM - is well-documented, widely supported, and appropriate for building scalable web applications. All tools used are open-source or freely available. The team has access to personal development machines capable of running the full stack locally. No specialized hardware or proprietary software licenses are required.

2.3.2 Operational Feasibility

The system is operationally feasible. The web-based nature of the platform means end-users require only a modern browser and internet connection - no software installation needed. The role-based interface (Customer, Employee, Admin) ensures each user type sees only relevant functionality, reducing the learning curve. The system mirrors workflows already familiar to utility staff (meter reading, bill generation, complaint handling) but automates and digitizes them. From a management perspective, the admin dashboard provides consolidated views of operations, reducing the need for manual reporting.

2.3.3 Economic Feasibility

The system is economically feasible. The entire development stack is built on open-source and free-tier technologies, resulting in zero software licensing costs. Development is carried out by the student team at no labor cost. For a real-world deployment by K-Electric, the operational cost would be minimal compared to the savings gained from reduced paper billing, fewer manual processes, and decreased customer service overhead. The long-term economic benefit significantly outweighs the one-time development investment.

2.4 Project Goals

FUNCTIONAL GOALS:

- Provide customers with online access to their bills, payment history, and usage analytics.
- Enable employees to record meter readings digitally and manage work orders.
- Allow administrators to generate bulk bills, manage tariffs, and oversee complaints and outages.
- Implement a formal complaint submission and tracking workflow.
- Provide a new electricity connection application and approval process.

TECHNOLOGICAL GOALS:

- Build the system using a modern, type-safe full-stack framework (Next.js + TypeScript).
- Use an ORM-based database layer (Drizzle ORM + MySQL) for maintainable, schema-driven data management.
- Implement secure role-based authentication using NextAuth with JWT session management.

QUALITY GOALS:

- Ensure data accuracy in billing through automated tariff slab calculations.
- Maintain secure access control so each role can only access authorized resources.
- Provide a responsive UI accessible on both desktop and mobile browsers.

ORGANIZATIONAL GOALS:

- Apply Agile/Scrum methodology to structure development into manageable sprints.
- Deliver the project within the 8-week academic timeframe with all core features functional.

CONSTRAINTS:

- Development is constrained to the remaining 8 weeks of the Spring 2026 semester.
- The system will not integrate with live K-Electric APIs or real payment gateways (simulated).
- IoT-based smart meter integration is out of scope.
- Mobile native applications (iOS/Android) are out of scope.
- System deployment to production is out of scope for this academic submission.

2.5 Project Abstract

The Electrolux Energy Management System (EMS) is a comprehensive web-based solution designed to modernize electricity utility operations in Karachi, Pakistan. Addressing the inefficiencies of K-Electric's current paper-based and manual processes, this system provides a centralized digital platform for customers, employees, and administrators. Customers gain transparent access to bills, payment tracking, usage analytics, and complaint management. Employees can digitally record meter readings and manage work orders, while administrators oversee operations through dashboards, bulk billing, tariff management, and outage coordination. Built using Next.js, TypeScript, MySQL, and Drizzle ORM, the system employs role-based authentication, automated billing calculations, and responsive design to deliver a secure, efficient, and user-friendly experience. Developed over 8 weeks using Agile/Scrum methodology, this project aims to bridge the gap between utility providers and consumers, setting a new standard for digital utility management in Pakistan.

2.6 Project Scope

IN SCOPE:

- Web-based customer portal with bill viewing, payment tracking, usage analytics, complaint submission, and new connection applications.

- Employee portal for digital meter reading recording, work order management, and reading request assignment.
- Administrator portal for customer/employee management, bulk billing, tariff management, complaint oversight, and outage scheduling.
- Automated tiered tariff slab calculation and bill generation engine.
- Secure role-based authentication and authorization using NextAuth with JWT.
- 20 core features as detailed in the Feature List section.

OUT OF SCOPE:

- Native mobile applications (iOS/Android).
- Real-time integration with physical smart meters or IoT devices.
- Online payment gateway integration (payments are recorded, not processed).
- SMS or email notification systems.
- Production deployment or hosting infrastructure setup.

2.7 Project Team

All three team members contribute equally across the full stack — including frontend development, backend API design, database schema, testing, and documentation. No member is restricted to a single role.

Name & Roll Number	Availability	Contribution
Huzaifa Abdul Rehman (23K-0782)	100%	Full-stack development, database, frontend, backend, testing
Muhammad Abdullah Khan (23K-0607)	100%	Full-stack development, database, frontend, backend, testing
Abdul Moiz Hussain (23K-0553)	100%	Full-stack development, database, frontend, backend, testing

2.8 Existing & Related Work

The table below compares Electrolux EMS against the existing K-Electric online portal and a generic international solution (SmartGrid Solutions):

Feature	K-Electric Current	SmartGrid (US)	Electrolux EMS
Online Bill Viewing	Limited	Yes	**Yes**
Bill Estimation Calculator	No	No	**Yes**
Usage Analytics / Graphs	No	Yes	**Yes**
Complaint Tracking Lifecycle	No	No	**Yes**
Zone-Based Outage Management	No	No	**Yes**
Meter Reading Request	No	No	**Yes**

New Connection Application	Partial	No	**Yes**
Employee Work Order Tools	No	Yes	**Yes**
Admin Bulk Bill Generation	No	No	**Yes**
Tiered Tariff Slab Calculation	No	No	**Yes**
Local Pakistan Focus	Yes	No	**Yes**

Uniqueness of Electrolux EMS: Unlike generic international solutions, Electrolux EMS is specifically tailored to address Karachi's infrastructure constraints, K-Electric's operational workflows, and the socio-economic realities of Pakistani consumers. It provides comprehensive role-based functionality in a single integrated platform, emphasizing transparency, accountability, and ease of use for all stakeholders.

2.9 Proposed Solution & Architecture

Electrolux EMS follows a modern three-tier web architecture with a clear separation of concerns:

PRESENTATION LAYER:

- Built with Next.js 14 and React 18, using Tailwind CSS for responsive styling. Provides role-specific dashboards and UI components for Customer, Employee, and Admin users.

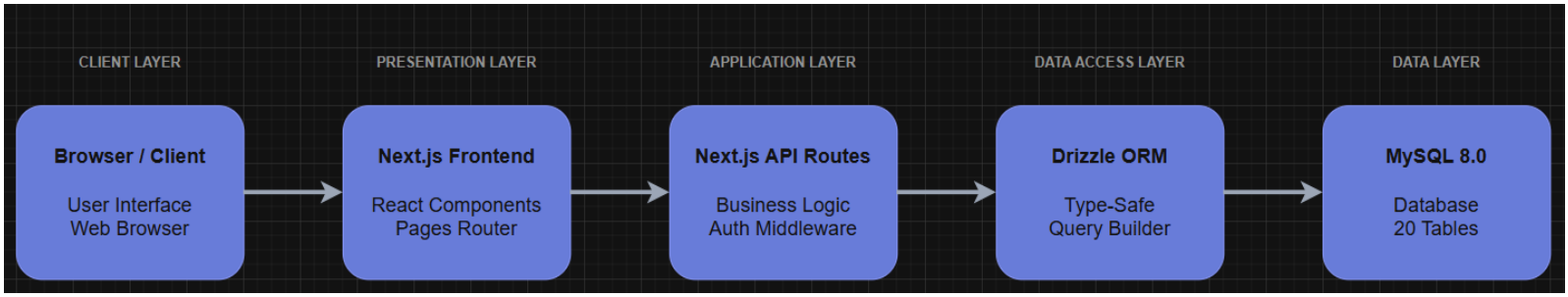
APPLICATION LAYER:

- Next.js API routes handle business logic, authentication (NextAuth with JWT), billing calculations, meter reading workflows, complaint management, and role-based access control.

DATA LAYER:

- MySQL 8.0 database managed through Drizzle ORM with a 20-table schema covering users, customers, employees, bills, payments, meter readings, complaints, outages, tariffs, and more.

Architecture Flow:



2.10 Use Case Model

See attached link:
https://drive.google.com/drive/folders/1OSXGavwJ1W7aPo7mCAhk9eMlsT54XNGg?usp=drive_link

The system involves three primary actors: Customer, Employee, and Administrator. Key use cases per actor:

CUSTOMER:

- View/download bills, estimate bill, track payments, view usage analytics, submit complaints, request meter reading, apply for new connection, manage account.

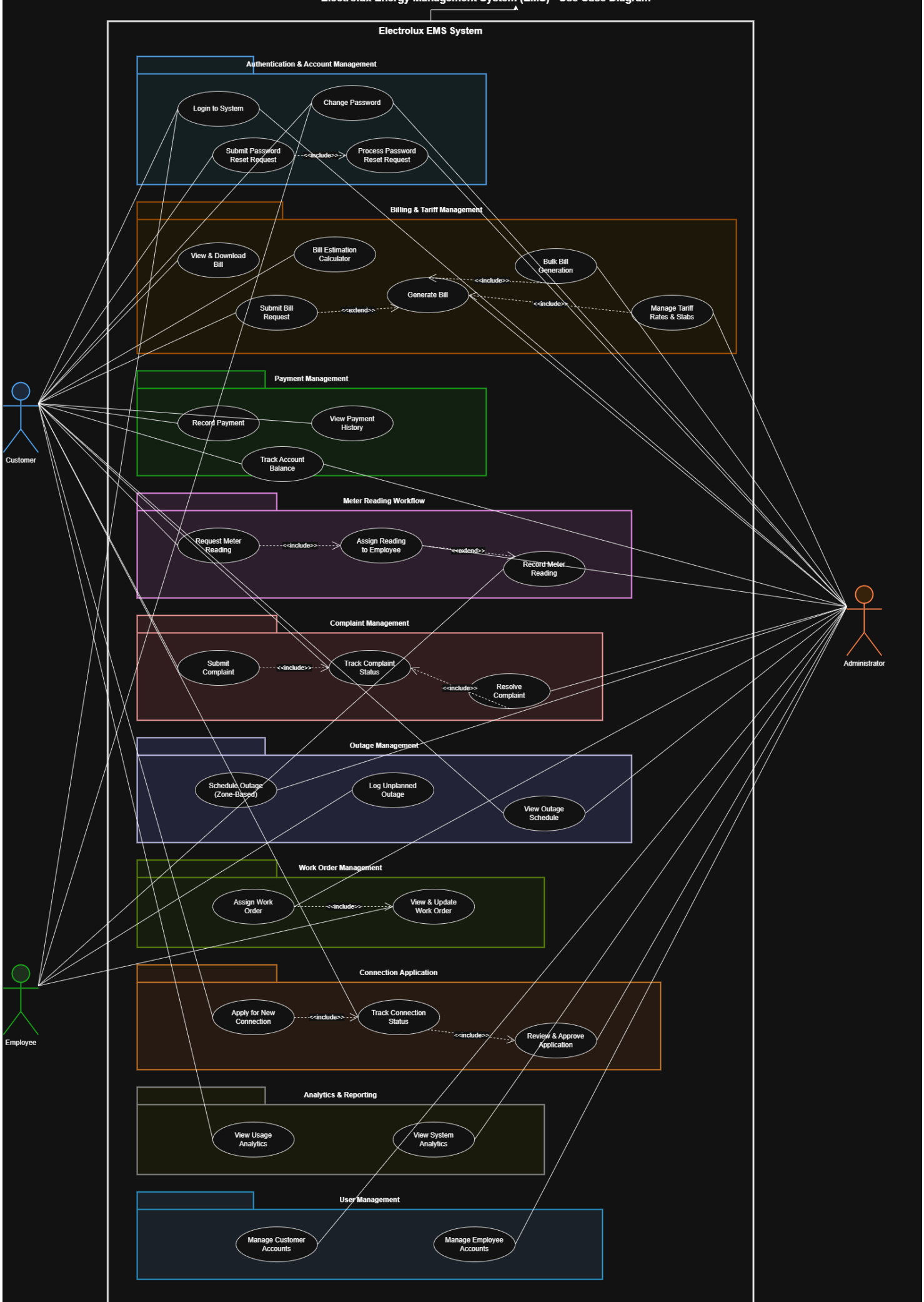
EMPLOYEE:

- Record meter readings, view assigned reading requests, manage work orders.

ADMINISTRATOR:

- Manage customers/employees, generate bulk bills, configure tariffs, resolve complaints, schedule outages, assign reading requests, view system analytics, manage password reset requests.

Electrolux Energy Management System (EMS) - Use Case Diagram



2.11 Feature List

The following 20 core domain features define the functional scope of Electrolux EMS. Login/logout and dashboards are excluded as generic features.

- **Role-Based Portal System:** Dedicated web portals for Customer, Employee, and Admin roles with JWT-based session management and route-level access control.
- **Electricity Bill Generation:** Automatic generation of monthly electricity bills based on meter readings and applicable tariff.
- **Tiered Tariff Slab Calculation:** Automated calculation of charges using progressive slab-based tariff rates per customer category.
- **Bulk Bill Generation with Preview:** Admin generates bills for all customers in a single operation with preview and confirmation.
- **Online Bill Viewing & Download:** Customers view detailed bills with slab breakdown, taxes, due dates and download/print them.
- **Bill Estimation Calculator:** Customers estimate their bill by entering units consumed, calculated against current tariff slabs.
- **Payment Recording & Tracking:** Record payments against bills with automatic status updates (paid/unpaid/overdue) and full history.
- **Meter Reading Recording by Employee:** Field employees digitally submit meter readings for assigned customers.
- **Meter Reading Request by Customer:** Customers formally request a meter reading through the portal, initiating an assignment workflow.
- **Reading Request Assignment to Employee:** Admins assign pending meter reading requests to specific field employees for fulfillment.
- **Employee Work Order Management:** Employees view, update, and complete assigned work orders.
- **Customer Management (Admin):** Admins create, view, edit, and deactivate customer accounts.
- **Employee Management (Admin):** Admins manage employee accounts and work assignments.
- **Tariff Rate Management:** Admins define and update electricity tariff categories and slab rates.
- **Customer Usage Analytics:** Visual charts showing monthly consumption trends and cost patterns for customers.
- **Administrative Analytics & Reporting:** System-wide analytics covering revenue, consumption by zone, complaint volumes, and outage stats.
- **Complaint Submission & Tracking:** Customers submit complaints and track resolution through a formal lifecycle: submitted → in progress → resolved.
- **Zone-Based Outage Management:** Outages are scheduled, recorded, and tracked by geographic zone.
- **New Connection Application & Tracking:** Customers apply for a new electricity connection and track approval status.
- **Password Reset Request & Tracking:** Customers submit password reset requests through the portal; admins track and process them.

2.12 Software Development Methodology

Electrolux EMS will be developed using the **Agile Software Development Methodology**, specifically following the **Scrum framework**. The feature set is decomposed into user stories grouped into 2-week sprints. Regular sprint reviews allow early detection of issues. All three team members participate equally in planning, development, and review activities.

Sprint Structure (4 sprints, 2 weeks each, 8 weeks total):

Sprint 1 (Weeks 1-2):

- Project setup, repository initialization, database schema (20 tables), authentication system, role-based portal base structure.

Sprint 2 (Weeks 3-4):

- Bill viewing & download, bill calculator, payment tracking, usage analytics, billing engine (tariff slab calculation), bulk bill generation.

Sprint 3 (Weeks 5-6):

- Meter reading recording, meter reading requests & assignment, employee work orders, customer/employee management (admin), tariff management, new connection application & tracking.

Sprint 4 (Weeks 7-8):

- Complaint submission & tracking, zone-based outage management, bill request workflow, password reset & tracking, account balance & arrears, system-wide testing, documentation.

2.13 Proposed Infrastructure, Platform and Tools

PROGRAMMING LANGUAGES:

- TypeScript, JavaScript, SQL

FRAMEWORKS & LIBRARIES:

- Next.js 14, React 18, Tailwind CSS, Drizzle ORM, NextAuth v4, Chart.js / React-ChartJS-2, Zod, React Hook Form, bcryptjs, Framer Motion, Lucide React, Axios, date-fns

DATABASE:

- MySQL 8.0

DEVELOPMENT TOOLS:

- Visual Studio Code, Git, GitHub, MySQL Workbench, Drizzle Studio, Postman, Node.js 18+, npm

DESIGN & DOCUMENTATION:

- draw.io / diagrams.net (use case diagrams), ProjectLibre (Gantt chart)

2.14 Project Timeline

8-week sprint-based timeline summary for Spring 2026:

Week 1-2 (Sprint 1):

- Project setup, repository, DB schema (20 tables), authentication, role-based portal structure.

Week 3-4 (Sprint 2):

- Bill viewing & download, bill calculator, payment tracking, usage analytics, billing engine, bulk bill generation.

Week 5-6 (Sprint 3):

- Meter reading recording, reading requests & assignment, work orders, customer/employee management, tariff management, connection application.

Week 7-8 (Sprint 4):

- Complaint tracking, outage management, bill request workflow, password reset, account balance, system testing, documentation.

2.15 Mockups (Optional)

Not included in this submission.

2.16 References

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GITHUB REPO: https://github.com/HuzaifaAbdulRehman/electrolux_ms

END OF PROPOSAL